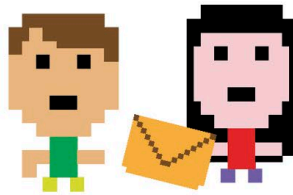


How it works

The program gets the information about capital cities from a text file. You'll use the **Tkinter** module to create the pop-up boxes that let the program and user communicate. When a new capital city is entered by a user, the information is added into the text file.

► Communication

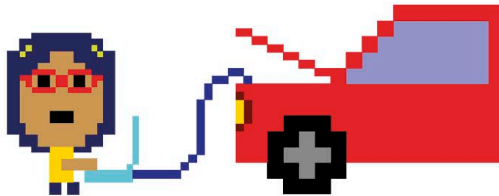
The program uses two new **Tkinter** widgets. The first, **simplifiedialog()**, creates a pop-up box that asks the user to input the name of a country. The second, **messagebox()**, displays the capital city.



■ ■ LINGO

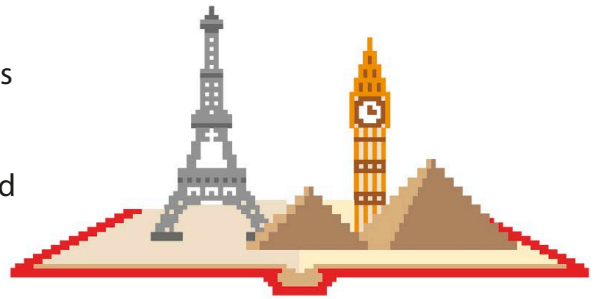
Expert systems

An expert system is a computer program that is a specialist on a particular topic. Just like a human expert, it knows the answers to many questions, and it can also make decisions and give advice. It can do this because a programmer has coded it with all the data it needs and rules about how to use the data.



△ Auto wizards

Motor companies create expert systems that are full of information about how their cars function. If your car breaks down, a mechanic can use these systems to solve the problem. It's like having a million expert mechanics look at the problem rather than just one!



△ Dictionaries

You'll store the names of countries and their capitals in a dictionary. Dictionaries work a bit like lists, but each item in a dictionary has two parts, called a key and a value. It's usually quicker to look things up in a dictionary than it is to find something in a long list.

▽ Ask the Expert flowchart

When the program starts, it reads the data from a text file. It then uses an infinite loop to keep asking questions, and only stops when the user quits the program.

